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B(1st Sm.)-Physics-H/DSCC-1/CCF

2024

PHYSICS — HONOURS

Paper : DSCC-1

(Basic Physics - I)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question no.* 1** and ***any five*** questions, taking ***at least one*** from *each* ***Group***.

1. Answer ***any five*** questions : 3×5

(a) Sketch the functions $f(x)=|x|$ and $g(x)=\frac{df}{dx}$, in the range $-2 \le x \le +2$. Comment on the continuity and differentiability of the given functions.

(b) Construct a differential equation of the form $y''(x)+ay'(x)+by(x)=0$, where '′' indicates derivative with respect to x and a, b are constants, for which the solutions are e^{2x} and e^{-2x}.

(c) Evaluate the integral $\iint_S \sqrt{x^2+y^2}\,dxdy$ (by transforming to plane polar coordinates or otherwise), where S is a region in the x-y plane bounded by $x^2+y^2=4$ and $x^2+y^2=9$.

(d) Find the first three non-zero coefficients in the Taylor series expansion of $\frac{1}{\cos x}$ about $x=0$.

(e) A particle is moving in 3-dimensions under a conservative force field. Show that the sum of its kinetic and potential energies is conserved.

(f) A mass m with speed v approaches a stationary mass M. The masses bounces off each other. Find the initial velocities of the particles in the CM frame. Assume that the motion takes place in 1-dimension.

(g) A system consists of two particles of masses m_1 and m_2. Prove that the centre of mass divides the line joining m_1 and m_2 into two segments whose lengths are in the ratio $m_2 : m_1$.

(h) What are streamlines? Explain why two streamlines can never intersect.

Group - A

2. (a) If $z=\ln\sqrt{x^2+y^2}$, show that $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=1$.

(b) Show that $df=y(1+x-x^2)dx+x(x+1)dy$ is not an exact differential. Find out a suitable integrating factor so that the differential becomes an exact differential.

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(c) Plot $r = a(1 - \cos\theta)$ in polar coordinates (r, θ) in the range $0 \le \theta \le 2\pi$, where a is a constant.

(d) Solve the differential equation $x''(t) - x(t) = 0$, with the boundary conditions $x(0) = 1$ and $x(\infty) = 0$. Here '' indicates derivative with respect to t. 3+4+2+3

3. (a) Using the summation convention and Levi-Civita symbol, show that $\vec{\nabla} \times \vec{\nabla}\phi = 0$, where $\phi(x, y, z)$ is any scalar field.

(b) Find the flux of the vector $\vec{F} = z\hat{K}$ over a sphere of radius a centered at origin.

(c) For a scalar field $\Phi(\vec{r})$, show that $\vec{\nabla}\Phi(\vec{r})$ is always normal to the Φ = constant surface at $\vec{r}$. Find a unit vector normal to the surface $x^2 + y^2 + z^2 = 3$ at (1,1,1).

(d) Given $\vec{F} = z^2\hat{i} + xy\hat{j} + x^2\hat{k}$, find $\nabla^2\vec{F}$. 3+3+(2+2)+2

4. (a) Find the line integral of the vector $\vec{F} = x^2\hat{i} + y^2\hat{j} + (xz - y)\hat{k}$ along the straight line joining the points (0, 0, 0) to (2, 3, 4).

(b) Solve the differential equation $y'' - 5y' + 6y = 2\cos x$.

(c) Find the expression for $\vec{\nabla}$ in cylindrical coordinate system. 3+5+4

Group - B

5. (a) A particle of mass m, moving in a horizontal line with initial velocity v_0 is acted upon by a resistive force γv^2 (γ is constant).

(i) Set up the differential equation of motion of the particle.

(ii) Find the speed $v(t)$.

(iii) Plot $v(t)$ against t.

(b) The coordinates of a particle moving in the x-y plane are given by $x(t) = at$ and $y(t) = bt^2$, where $a = 2$ m/s and $b = 0.5$ m/s^2. Find the velocity and acceleration of the particle at $t = 3$ Sec. Find the equation of the trajectory of the particle. (1+3+2)+(4+2)

6. (a) A particle of mass m and total energy E moves along the x-axis in a potential $V(x)$, starting from x_0 at time t_0. If the particle reaches x at time t, show that

$$t - t_0 = \pm \int_{x_0}^{x} dx' \frac{1}{\sqrt{\frac{2}{m}[E - V(x')]}}.$$

(b) A particle of mass m moves in a potential field $V(x) = m(x^3 - 3x)$.

(i) Sketch the potential $V(x)$ against x.

(ii) Find the positions of stable and unstable equilibrium.

(iii) Find the force acting on the particle.

(iv) If the particle is displaced slightly from stable equilibrium, then find the frequency of small oscillation. 4+(2+3+1+2)

7. (a) Show that if the total linear momentum of a system of particles is zero, the angular momentum of the system is independent of the choice of origin.

(b) Show that for a system of particles $\frac{d\vec{L}}{dt} = \vec{N}$, where $\vec{L}$ is the total angular momentum and $\vec{N}$ is the total external torque (assume internal forces are pairwise central forces).

(c) Find out the gravitational potential at a point inside a uniform solid sphere. 3+5+4

8. (a) For a particle of mass m moving in a central potential $U(r)$, show the energy is given in polar coordinates by $E = \frac{1}{2}m\dot{r}^2 + U_{eff}(r)$, where $U_{eff}(r) = U(r) + \frac{\ell^2}{2mr^2}$, ℓ being the angular momentum.

(b) A particle of mass m moving in a central force field describes a spiral $r = k\theta^2$, where k is a positive constant.

(i) Find the force law.

(ii) Find the total energy of the system.

(c) Consider a head-on elastic collision in one dimension between a heavy mass (m_1) and a light mass (m_2) ($m_1 >> m_2$), where m_2 is initially at rest. Show that after the collision, the light mass rebounds with a speed equal to about twice the initial speed of m_1. 3+(2½+2½)+4

9. (a) Write down the equation of continuity for a moving fluid. Explain the physical significance of this equation.

(b) Establish that the condition of equilibrium of an incompressible fluid under body force $\vec{F}$ per unit mass is given as $\vec{F} = \frac{1}{\rho}\vec{\nabla}P$ when P is the pressure at some point (x, y, z) and ρ is the density of the fluid. Hence establish Pascal's law.

(c) Using Bernoulli's theorem for an incompressible fluid, show with suitable example that high velocity point on a streamline corresponds to low pressure and vice versa. 3+(4+2)+3